

Sahil Gajera

Worcester, USA,
+1(609)-369-5534

shahilgajera12@gmail.com

Summary

Graduate student in Robotics and Automation with specialized expertise in Computer Vision, Motion Planning, and Robot Control Systems. Currently contributing to autonomous vehicle research, focusing on LiDAR-camera sensor fusion and terrain segmentation for UGV navigation. Proficient in ROS2, MATLAB, Python, and C++, with a strong foundation in robot state-space representation, kinematics, and dynamics. Experienced in developing perception pipelines for real-time autonomous systems and mentoring robotics teams. Seeking opportunities to apply technical skills in autonomous robotics and intelligent systems development.

Experience

Volunteer Research Assistant

AVMI (Autonomous Vehicle Mobility Institute), Worcester, USA

Sep 2025 — Current

Contributing to MassTech off-road autonomous UGV project focusing on perception and simulation integration. Developing LiDAR-camera data fusion pipeline for semantic segmentation and environment understanding in outdoor terrains. Working with ROS2-Unreal Engine integration under Ubuntu for real-time simulation and data management. Implementing terrain segmentation techniques and geometry for autonomous navigation in unstructured environments. Applying computer vision, perception, and simulation frameworks in robotics research.

Skills: Computer Vision, ROS2, Python, LiDAR-Camera Fusion, Semantic Segmentation, MATLAB, Ubuntu.

Graduate Assistant (Teaching Assistant)

Worcester Polytechnic Institute, Worcester, USA

Oct 2025 — Current

Assist course instructors in managing academic and administrative aspects of graduate-level engineering courses. Responsibilities include organizing course materials, coordinating student communications and helping maintain consistency in grading and assessment.

Robotics Competition Trainer

RFL Academy, Mumbai, India

Oct 2023 — May 2025

Mentored students in robotics competitions (FRC, FTC, WRO), teaching programming in Java and Python for autonomous and tele-operated robots. Developed computer vision pipelines for object detection and tracking. Integrated control systems and networking for multi-sensor platforms using motor controllers, encoders, IMUs, and vision modules on embedded systems (Raspberry Pi, Lego, Linux-based).

Skills: Java, Python, Computer Vision, Control Systems, Embedded Systems, Mentoring

Education

Master of Science, Robotics And Automation, Worcester, MA, 2025 — 2027

Worcester Polytechnic Institute

Bachelor of Technology, Information Technology, Rajkot, India, 2020 — 2024

RK University

GPA/CGPA 8.99